

1. Project Overview

1.1 The Vision

Opal is a **Social Operating System (SOS)** that seeks to modernize how people communicate and plan. Its main features include:

1. **Resonance Messaging**
 - Voice-based messaging capturing tone and emotion.
 - Support for text-to-voice or voice-to-text functionalities.
2. **Social Flow Calendar**
 - An “invisible” scheduling system—no rigid calendar grids.
 - Create and manage events seamlessly with friends or social circles.
3. **AI Agents & Translation**
 - AI-driven assistants that can help with scheduling, tasks, or conversation flow.
 - Translation features to facilitate multilingual interactions.
4. **Social Circles**
 - Organized group management (e.g., Family, Close Friends).
 - Pre-defined circles plus customizable color-coding/icons.

Q1 Goal (MVP):

- **Deliver** both the **Elixir backend** and a **local-storage-enabled** frontend for basic offline usage.
 - Capture essential user data (messages, events, etc.).
 - Lay the groundwork for advanced AI features in subsequent iterations.
-

2. Team Structure

2.1 Current Team Members

- **Frontend (React Native Expo)**
 - **Marco**
 - **Sam**
 - **Jaka**
- **Backend (Elixir)**
 - Two Elixir developers will join soon (TBA) to build and deliver the MVP backend in Q1. Database: ScyllaDB.
- **AI/NLP**
 - One engineer responsible for AI agents, translation, voice models (on hold until core MVP is ready).
- **Design**
 - **Folusho**: Manages UI/UX, wireframes, visual branding.

2.2 Responsibilities Overview

- **Front End Team**
 - **Status:** The fundamental UI is built with **React Native (Expo)**.
 - **Focus:**
 1. Implementing core functionalities (messaging, event flows, circle management).
 2. Integrating **local storage** (e.g., SecureStore, SQLite) for offline availability.
 3. Preparing to sync data with the upcoming Elixir backend.
 - **Backend Team (Elixir)**
 - Will manage user authentication, event scheduling, and message persistence.
 - Delivers real-time (Phoenix Channels) or REST/GraphQL endpoints for the app.
 - **Target:** A working backend by end of Q1 for the MVP launch.
 - **AI/NLP Engineer**
 - Develop advanced features: AI-driven agents, translation, voice-based interactions.
 - Integrate with the Elixir backend once MVP fundamentals (auth, data flow) are stable.
 - **Design (Folusho)**
 - Finalize wireframes, user flows, color palettes, icons in Figma.
 - Provide updated UI elements as features evolve.
-

3. Repositories & Codebase

All projects are housed under **NIOV-Labs** on GitHub:

- **Frontend**
 - **Repo:** [Opal](#)
 - **URL:** github.com/NIOV-Labs/Opal
 - Holds the **React Native (Expo)** code, local storage logic, and offline features.
- **Backend**
 - **Repo:** [OpalBackend](#)
 - **URL:** github.com/NIOV-Labs/Opalbackend
 - Elixir/Phoenix services for user auth, messaging, event management (in development for Q1).
- **AI/NLP**
 - **Status:** Will be created once the AI engineer is actively building agent logic and translation features.

(Maps or external data sharing are **not** part of the scope.)

4. Development Methodology

We use a **semi-Scrum** approach, with daily coordination:

- **Daily Stand-ups**
 - **Duration:** 15–30 minutes.
 - **Purpose:** Share progress, highlight blockers, assign tasks.
 - **Attendees:** Frontend devs, upcoming Elixir devs, AI engineer, and designer.
 - **Task Management (ClickUp)**
 - **Board & Backlog:** All user stories, tasks, bugs stored here.
 - **Assignment:** Decided in stand-ups or short planning meetings.
 - **Progress Tracking:** Team updates tasks in real time for transparency.
 - **Sprint-like Cycles**
 - Usually **1–2 weeks**.
 - Each cycle aims for a functioning increment (resonance feature, calendar flows, etc.).
 - **Retrospectives** (Optional / As Needed)
 - Discuss successes and bottlenecks to refine the process continually.
-

5. Current Focus & Roadmap

5.1 Current Status

- **Frontend**
 - **UI Foundation** in React Native is ready.
 - Actively implementing **functionality**: local storage for offline messages, events, contact management.
- **Backend**
 - Elixir devs expected to arrive soon; scoping initial MVP features.
 - By the end of Q1, we aim to have a stable backend delivering minimal user auth, data persistence (for messages, events, circles), and optional real-time channels.
- **AI/NLP**
 - Basic conceptual design is in place, but actual AI integration will ramp up **after** the Q1 MVP foundation (backend + local storage) is stable.
- **Design**
 - Figma prototypes are mostly done for the MVP.
 - Folusho will provide iterative updates for new or refined features.

5.2 Q1 Priorities (MVP)

1. **Resonance Messaging**
 - Voice & text messaging with basic tone replication placeholders.
 - Offline storage (so users can still view or send messages without connectivity, to be synced with the backend later).

2. **Social Flow Calendar**

- Simple event creation, invites, RSVP flows.
- Local storage for offline usage, with eventual sync to the Elixir backend.

3. **AI Agents & Translation (Placeholder)**

- UI and architecture planning.
- Core AI features likely introduced after baseline messaging and calendar are stable.

4. **Social Circles**

- Implement circle management with color-coding/icons.
- Local data first; back-end sync once Elixir endpoints are ready.

5. **Privacy & Data Storage**

- Ensure local storage is secure (e.g., SecureStore).
- Plan for data anonymization or minimal necessary data storage if needed.

5.3 Beyond Q1

● **AI Agents**

- Develop interactive agents for user tasks or auto-translation.
- Integrate with the Elixir backend to manage logic and data flows.

● **Enhanced Offline/Online Sync**

- Smooth synchronization when users go back online (messages, events, circle changes).

● **Extended Privacy Controls**

- Let users set voice permissions or translation usage disclaimers.
- Offer advanced encryption or user-consent flows if needed.

5.4 High-Level Roadmap Timeline

1. **Month 1**

- **Design & Figma Completion**
 - i. Folusho finalizes all wireframes and design elements.
 - ii. Any remaining design tweaks or adjustments are addressed.
- **Initial Expo React Native UI**
 - i. Core screens and navigation are created in React Native (Expo).
 - ii. Basic structures for Resonance Messaging, Social Flow Calendar, AI placeholders, and Social Circles are in place.

2. **Month 2**

- **Finalize UI & Add Functionality**
 - i. Complete any remaining UI components.
 - ii. Implement **offline storage** (e.g., SecureStore/SQLite) for messages, contacts, and event data.
 - iii. Build out **onboarding flows** (account setup, basic user preferences).
- **User Flows & Testing**
 - i. Ensure smooth transitions between screens and handle edge cases in offline scenarios.
 - ii. Conduct preliminary QA to validate the offline-first approach.

3. **Month 3**

- **Elixir Backend Endpoints**

- i. The two Elixir devs establish core services (authentication, event management, messaging).
 - ii. Decide on real-time (Phoenix Channels) or REST/GraphQL endpoints as needed.
- **Sync with Frontend**
 - i. Integrate the React Native app's offline data with the backend—allowing seamless **online/offline** usage.
 - ii. Begin handling user authentication, data syncing (messages, events, circles), and any real-time features.
- **Refinement & QA**
 - i. Perform end-to-end testing to confirm that offline and online data remain consistent.
 - ii. Address performance, security, and UX improvements for the MVP launch.

(Exact dates may vary based on onboarding speed and new requirements.)

6. Quick Notes & Next Steps

- **Collaboration**
 - **Frontend**: Continue building out offline functionalities.
 - **Backend**: Elixir devs set up essential endpoints (auth, messages, events) as soon as they join.
 - **AI**: Work to start once the MVP core is stable.
 - **Communication**
 - **ClickUp** for tasks and backlog management.
 - **Daily Stand-ups** (15–30 mins) to ensure alignment.
 - **WhatsApp/Discord** for quick chats.
 - **Documentation**
 - This doc is your **quick-start reference**.
 - Each repo will include more technical details in its [README.md](#) or Wiki as the project evolves.
-

7. Welcome Aboard!

We're delighted to have you on the **Opal** journey. If you have any questions, feel free to reach out via **daily stand-ups**, **WhatsApp**, or **Discord**. Together, we'll deliver a next-generation social experience that seamlessly blends **offline support**, **real-time Elixir backend**, and eventually **AI-driven interactions**.

Last Updated: 1.3.2025